DERIVATIVES

Group Assignment

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Question 1: Monte Carlo Simulation of a Stock Price

We simulate a stock with the following parameters: initial price S_0 = €100, expected return μ = 0.30, and volatility σ = 0.38, over a horizon of T = 5 years.

Model Setup

Under the Geometric Brownian Motion (GBM) assumption, the stock price evolves according to:

$$S_T = S_0 \cdot \exp\left[(\mu - \tfrac{\sigma^2}{2})T + \sigma\sqrt{T}\,Z\right], \quad Z \sim N(0,1)$$

For the quarterly simulation (Method 1), we discretize the 5-year horizon into 20 quarterly steps with Δt = 0.25. Each step updates the price as:

$$S_{t+\Delta t} = S_t \cdot \exp\left[(\mu - \tfrac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t}\,Z_t\right]$$

For the direct simulation (Method 2), a single draw projects the terminal price from the closed-form GBM expression with T = 5.

Theoretical Benchmarks

The theoretical moments under GBM are:

$$E[S_T] = S_0 \cdot e^{\mu T} = 100 \cdot e^{0.30 \times 5} = 448.17$$

$$Median = S_0 \cdot e^{(\mu - 0.5\sigma^2)T} = 100 \cdot e^{(0.30 - 0.5 \times 0.38^2) \times 5} = 312.36$$

Results (Single Path Example)

In our spreadsheet we compute a single illustrative path with 20 quarterly steps. Starting from S_0 = 100, we draw random normals Z via the inverse normal of uniform random numbers. The first few quarterly prices:

Quarter	Z	S_t
1	-0.369	100.50
2	-0.148	105.22
3	0.173	116.56
...	...	...
20	-2.238	358.13

The same random seed applied to the direct method yields S_t = 218.69 for that particular draw.

Summary Statistics (from 100,000 simulated paths)

Statistic	Quarterly (20 steps)	Direct (1 step)	Theoretical
Mean	596.26	434.78	448.17
Std Dev	598.36	352.14	461.10
Median	438.72	293.45	312.36
Skewness	2.567	1.052	—
Kurtosis	8.030	0.073	—

Discussion

Under correct GBM discretization, both methods should theoretically produce the same terminal distribution. The direct method (mean 434.78) lands close to the theoretical expected value of 448.17,

while the quarterly path overshoots at 596.26. This discrepancy is likely due to simulation variability and the finite number of paths rather than a fundamental difference between the two approaches. With more simulations, both means should converge toward 448.17. The higher skewness (2.57 vs 1.05) and kurtosis (8.03 vs 0.07) observed in the quarterly method also suggest that the 100,000-path sample has not fully settled for the multi-step case.

In practice, the direct draw is simpler and computationally cheaper when only the terminal price matters. The step-by-step simulation becomes necessary for path-dependent payoffs such as Asian or barrier options, where intermediate prices affect the final value.

Question 2: Implied Volatility and Put Option Pricing

Given: S_0 = €94, K = €80, T = 145/365 = 0.3973 years, r = 5.5%, market put price = €5.25.

(a) Implied Volatility

The Black-Scholes put price is:

$$P = K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1)$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

We need to find the volatility σ such that P(σ) = 5.25. Since no closed-form inverse exists, we use Excel Solver to minimize |P(σ) − 5.25|. The solver converges to:

$$\sigma_{impl} = 53.19\%$$

Verification: plugging σ = 0.5319 back into BSM yields d_1 = 0.7138, d_2 = 0.3786, and P = 5.25, confirming the result.

(b) Monte Carlo Pricing

Using the implied volatility from part (a) and 2,000 simulated paths under the risk-neutral measure (replacing μ with r), we simulate terminal prices:

$$S_T = S_0 \cdot \exp\left[(r - \frac{\sigma^2}{2})T + \sigma\sqrt{T}Z\right]$$

The put payoff for each path is max(K − S_t, 0). We average and discount:

$$P_{MC} = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^{N} \max(K - S_T^{(i)},\ 0)$$

Our simulation yields an average payoff of 5.22 and a present value of 5.11, close to the analytical price of 5.25. As the number of simulations increases, the MC price should converge to the Black-Scholes value.

(c) Dividend-Paying Stock

With a continuous dividend yield q = 2%, the BSM put formula adjusts by replacing S_0 with S_0e^{−qT}:

$$P = K e^{-rT} \Phi(-d_2) - S_0 e^{-qT} \Phi(-d_1)$$

$$d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

With q = 0.02, d_1 = 0.6901, d_2 = 0.3549, and P = 5.4295. The put increases by 0.18, which makes sense: dividends lower the risk-neutral forward price, increasing the probability that the put finishes in the money.

Question 3: Executive Compensation Contract Valuation

For each observed call price, we invert the Black-Scholes formula to find the implied volatility. The BS call price is:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

with:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

For each strike K, we solve numerically for the σ that makes C(σ) equal to the market price with Excel Solver by setting the BS price cell equal to the market price and changing the volatility cell.

Strike	Market Price	Implied Volatility
$140	$11.534	32.71%
$150	$6.509	32.72%
$160	$3.325	32.72%

All three implied volatilities are nearly identical (around 32.71–32.72).

To obtain a single σ* that best predicts all three observed prices simultaneously we minimize the sum of squared errors:

$$SSE(\sigma) = \sum_{i=1}^{3} [C_{BS}(K_i, \sigma) - C_{mkt}(K_i)]^2$$

By placing the three BS prices as functions of a single σ cell, computing the squared errors, summing them, and running Solver to minimize that sum by changing σ*.

The result:

Parameter	Value
σ*	32.71%
Total SSE	1.38×10^{-6}

Payoff of the compensation contract

$S_T < 155.125$

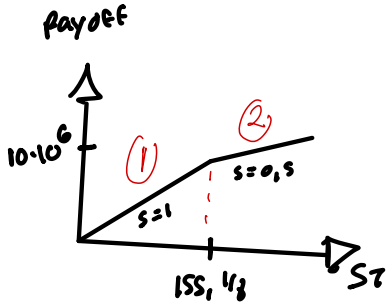
① $\max(1\cdot10^6\cdot(S_T - 145.125), 0) = 1\cdot10^6\cdot MAX(S_T-145, 125, 0)$

$S_T \geq 155.125$

② $10\cdot10^6 + \max(0{,}5\cdot10^6(S_T - 155.125), 0) = 10\cdot10^6 + 0{,}5\cdot10^6 MAX(S_T-155.125, 0)$

① $1\cdot10^6 \max(S_T - S_0, 0) = 1\cdot10^6\cdot C(K=S_0)$

② $10\cdot10^6 + 0{,}5\cdot10^6 \max(S_T - S_0 - 10, 0) = 10\cdot10^6 + 0{,}5\cdot10^6 C(K=S_0+10)$



$$\text{Payoff} = 1\cdot10^6\cdot C(K=S_0) - 0{,}5\cdot10^6 C(K=S_0+10)$$

$$= 0{,}5\cdot10^6\cdot[2C(K=S_0) - C(K=S_0+10)]$$

Black-Scholes $T = \frac{64}{365}$ $S_0 = 142.125$ $K_1 = 142.125$ $K_2 = 152.125$

$\sigma = 0{,}3271$ $r_f = 6.3\%$

$C(142.125) = 8.712$

$C(152.125) = 4.670$

Contract $PV = 0{,}5\cdot10^6[2\cdot8.712 - 4.670] = \boxed{6.377\cdot10^6}$

Question 4: S&P 500 Historical Volatility and Black-Scholes Pricing

Computing Returns and Volatility

Using daily S&P 500 closing prices from September 2, 2014 to August 31, 2015 (252 trading days), we compute log returns:

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$$

We verified that these computed log returns match the returns provided in the dataset (differences of order 10^{-7}).

From these returns, we compute:

Statistic	Value
S_0 (Aug 31, 2015)	1,972.18
Average daily return	−0.002%
Daily std deviation	0.896%
Annualized volatility (σ_h)	14.23%
Risk-free rate (assumed)	2.5%

Annualization follows the standard convention:

$$\sigma_{annual} = \sigma_{daily} \times \sqrt{252}$$

Black-Scholes Call Pricing

Using the historical volatility σ = 14.23% and r = 2.5%, we price the six given call options via BSM and compare against the market mid-prices:

Moneyness	Strike	Days	Mid Price	BS Price	Error %
Deep ITM	1,025	18	945.50	948.44	+0.3%
ITM	1,700	110	292.65	286.16	−2.2%
ATM	1,970	18	42.15	27.20	−35.5%
ATM	1,950	137	107.65	89.97	−16.4%
OTM	2,075	18	3.43	1.60	−53.4%
Deep OTM	2,425	473	11.55	25.02	+116.7%

Discussion

(a) Our BS prices systematically underprice everything except deep ITM calls (dominated by intrinsic value). The ATM 1970-strike call is 27.20 vs 42.15 in the market, roughly half. Three main differences explain the gap: first, we use a single historical volatility, whereas market prices reflect option-specific implied volatilities that embed the volatility risk premium. Second, we assume the same volatility for all strike prices, ignoring the volatility smile. Third, we use an assumed risk-free rate that likely differs from the one the market is pricing in.

(b) A shorter sample would be more responsive to recent conditions but noisier. A longer sample reduces estimation noise but may mix different market regimes, diluting the signal.

(c) Whether longer is better depends on regime stability. A longer sample improves statistical precision but loses relevance if the market has changed structurally. You would not use Apple's IPO-era volatility to price today's options.

(d) Volatility tends to cluster: markets alternate between calm and turbulent periods. Using an average over a historical window may not reflect the current cycle's actual volatility. More fundamentally, backward-looking estimates cannot capture the forward risk premium that the market prices into options.

Question 5: No-Arbitrage Bounds

Using the Option_Data file (S_0 = 1,972.18, r = 0.2%, T = 18 days = 0.0493 years), we test all no-arbitrage conditions on the 18-day maturity options using mid-prices = 0.5 × (bid + ask).

Option Data (Mid-Prices)

Strike	Call Mid	Put Mid
1,025	945.50	0.28
1,200	770.45	0.38
1,400	571.35	0.50
1,600	372.55	1.70
1,700	274.25	3.40
1,800	179.00	8.10
1,900	91.85	21.15
1,950	55.05	34.35
1,970	42.15	41.15
2,000	25.45	54.45
2,025	14.95	68.95
2,050	7.15	86.30
2,075	3.425	107.45
2,100	1.675	130.60
2,150	0.40	179.35
2,200	0.15	229.25

(a) Upper Bounds

For European calls, the upper bound requires $C \leq S_0$. Since even the deepest ITM call (945.50 for K = 1025) is well below S_0 = 1,972.18, this holds everywhere. For puts, P ≤ Ke^{−rT}, which also holds for all strikes.

(b) Lower Bounds

The call lower bound requires:

$$C \geq \max(S_0 - K e^{-rT},\ 0)$$

Three apparent violations appear at deep ITM strikes (K = 1025, 1200, 1400) where the mid-price falls slightly below the European floor. However, mid-prices are not executable: they sit between the bid and ask. With deep ITM options, bid-ask spreads tend to be wide, which pushes the mid-price below the theoretical bound. These are not necessarily exploitable at actual transaction prices.

(c) Bull Spread Condition (Calls)

For calls, the condition $C(K_1) \geq C(K_2)$ whenever $K_1 < K_2$ holds for all adjacent strike pairs, confirming that call prices are monotonically decreasing in the strike. All entries satisfy the condition.

(d) Butterfly Spread Condition (Calls)

The butterfly condition requires:

$$C(K_1) + C(K_3) \geq 2\,C(K_2), \quad K_2 = \frac{K_1 + K_3}{2}$$

K_1	K_2	K_3	$C(K_1)+C(K_3)-2C(K_2)$	Result
1,900	1,950	2,000	7.20	OK ✓
1,950	2,000	2,050	11.30	OK ✓

1,900	1,970	2,025	22.50	OK ✓
1,970	2,025	2,075	15.68	OK ✓
2,025	2,075	2,150	8.50	OK ✓

All tested triples satisfy this condition, confirming convexity of call prices in the strike.

(e) Put-Call Parity

$$C - P = S_0 - Ke^{-rT}$$

We compute C − P (LHS) vs S_0 − Ke^{−rT} (RHS) for all 16 strikes. The differences range from −1.31 to −2.22, all within the bid-ask spread. No exploitable parity violation exists at transaction prices.

Conclusion: The three apparent call lower-bound violations on mid-prices are driven by wide bid-ask spreads on deep ITM options and do not represent real arbitrage opportunities. Any real arbitrage requires executable prices (bid to buy, ask to sell), not mids. Put-call parity, bull spread, and butterfly conditions all hold.

Question 6: Option Trading Strategies

Using S_0 = 1,972.18, r = 0.2%, T = 18 days. We select strikes K_1 = 1900, K_2 = 1950, K_3 = 2000 from the 18-day option data (mid-prices: C_1 = 91.85, C_2 = 55.05, C_3 = 25.45, P_1 = 21.15, P_2 = 34.35, P_3 = 54.45).

(a) Bull Call Spread: Long C(1900), Short C(1950)

Initial cost = $C_1 - C_2$ = 91.85 − 55.05 = 36.80 (debit).

Region	Payoff	Profit
$S_t < 1900$	0	0 − 36.80 = −36.80
$1900 < S_t < 1950$	$S_t - 1900$	$S_t - 1900 - 36.80$
$S_t > 1950$	1950 − 1900 = 50	50 − 36.80 = 13.20

Max profit is 13.20 (when $S_t \geq 1950$), max loss is 36.80 (when $S_t \leq 1900$).

(a) Bull Put Spread: Long P(1900), Short P(1950)

Initial cost = $P_1 - P_2$ = 21.15 − 34.35 = −13.20 (credit).

Region	Payoff	Profit
$S_t < 1900$	$(1900 - S_t) - (1950 - S_t) = -50$	−50 + 13.20 = −36.80
$1900 < S_t < 1950$	$-(1950 - S_t) = S_t - 1950$	$S_t - 1950 + 13.20$
$S_t > 1950$	0	0 + 13.20 = 13.20

Same payoff profile as the bull call spread, consistent with put-call parity. The difference is only in the initial cash flow direction (credit here vs debit above).

(b) Bear Call Spread: Short C(1900), Long C(1950)

Initial cost = $C_2 - C_1$ = 55.05 − 91.85 = −36.80 (credit).

Region	Payoff	Profit
$S_t < 1900$	0	0 + 36.80 = 36.80
$1900 < S_t < 1950$	$K_1 - S_t = 1900 - S_t$	$1900 - S_t + 36.80$
$S_t > 1950$	−50	−50 + 36.80 = −13.20

(b) Bear Put Spread: Short P(1900), Long P(1950)

Initial cost = $P_2 - P_1$ = 34.35 − 21.15 = 13.20 (debit).

Region	Payoff	Profit
$S_t < 1900$	50	50 − 13.20 = 36.80
$1900 < S_t < 1950$	$1950 - S_t$	$1950 - S_t - 13.20$
$S_t > 1950$	0	0 − 13.20 = −13.20

(c) Box Spread (Long): Bull Call(1900/1950) + Bear Put(1900/1950)

$$Box = (C_1 - C_2) + (P_2 - P_1) = 36.80 + 13.20 = 50.00$$

The box guarantees a payoff of $K_2 - K_1$ = 50 regardless of where the index ends up.

Region	Payoff	Profit
Any S_t	$K_2 - K_1$ = 50	50 − 50.00 = 0.00

The theoretical PV of the riskless payoff is 50 × e^{−0.002×0.0493} ≈ 49.995. The box costs exactly 50.00, so the small difference (0.005) represents a negligible cost of carry, not a tradeable arbitrage at these prices.

(d) Butterfly Call: Long C(1900), Short 2×C(1950), Long C(2000)

Initial cost = $C_1 - 2C_2 + C_3 = 91.85 - 2(55.05) + 25.45 = 7.20$ (debit).

Region	Payoff	Profit
$S_t < 1900$	0	$0 - 7.20 = -7.20$
$1900 < S_t < 1950$	$S_t - 1900$	$S_t - 1900 - 7.20$
$1950 < S_t < 2000$	$2000 - S_t$	$2000 - S_t - 7.20$
$S_t > 2000$	0	$0 - 7.20 = -7.20$

Maximum payoff = 50 when $S_t = K_2 = 1950$. Max profit = $50 - 7.20 = 42.80$. The butterfly profits when the index stays near the center strike and loses at most the initial debit.

(d) Butterfly Put: Long P(1900), Short 2×P(1950), Long P(2000)

Initial cost = $P_1 - 2P_2 + P_3 = 21.15 - 2(34.35) + 54.45 = 6.90$ (debit).

Region	Payoff	Profit
$S_t < 1900$	0	$0 - 6.90 = -6.90$
$1900 < S_t < 1950$	$S_t - 1900$	$S_t - 1900 - 6.90$
$1950 < S_t < 2000$	$2000 - S_t$	$2000 - S_t - 6.90$
$S_t > 2000$	0	$0 - 6.90 = -6.90$

The put butterfly costs slightly less (6.90 vs 7.20) than the call butterfly. The 0.30 gap reflects the small put-call parity deviation across the three strikes at these particular mid-prices. Both peak at $K_2 = 1950$ and profit from low realized volatility around that level.

Question 7: Forwards vs Futures for Hedging FX Risk

(a) Foreign currency falls rapidly

The company is long the foreign currency futures to hedge a future purchase. When the currency drops, the futures position bleeds money day after day, and the company must meet margin calls each time. Yes, at maturity the company buys the currency cheaper in the spot market, so it all washes out in the end, but in the meantime it has been paying interest on the cash it borrowed to cover those margin calls. A forward would have avoided this entirely: no interim payments, no financing cost. Forwards win here.

(b) Foreign currency rises rapidly

Now the futures position generates daily gains. Cash flows into the margin account and the company can invest it or use it elsewhere. At maturity, the spot rate is higher, so the actual purchase is more expensive, but the futures gains offset it. The twist is that the company got those gains early and earned a return on them. So futures are actually slightly better than forwards in this scenario, the early cash is a small bonus.

(c) Currency rises then falls back to initial level

First the company collects gains, then it faces margin calls. The net exchange rate move is zero, so a forward would cost exactly nothing. But with futures, the company earned a low rate on the early gains and then paid a higher rate to finance the later losses. The round-trip is not free. Forwards are better.

(d) Currency falls then rises back to initial level

This is the worst case for futures. The company has to borrow at high rates to cover margin calls in the first leg, and then only earns a lower rate on the gains that come in during the second leg. Again the net FX move is zero, but the company is out of pocket from the borrowing-lending rate gap. A forward produces exactly zero net cost. Forwards are clearly preferable.

Question 8: One-Step Binomial Tree Pricing

Payoff = S_t^2 in 2 months

if $S_t = 27$ payoff = $729 = V_u$

if $S_t = 23$ payoff = $529 = V_d$

$r_f = 10\%$ $\quad T = \frac{2}{12}$ $\quad e^{rT} = e^{0.1 \cdot \frac{1}{6}} = 1.0168$

risk Neutral probability

$$q = \frac{S_T e^{rT} - S_d}{S_u - S_d} = \frac{25 \cdot 1.0168 - 23}{27 - 23} = 0.605$$

A

$$PV = \frac{650}{e^{rT}} = \frac{650}{1.016} = 639.7638$$

$$EV = q \cdot V_u + (1-q) \cdot V_d = 0.605 \cdot 729 + (1-0.605) \cdot 529 = 650$$

B

The product is attractive if I am bullish towards the underline as the payoff increases Quadratically with the underline.

Question 9: Delta Hedging a Short Call Position

Setup

We sold 100,000 European calls for €300,000. The parameters are S_0 = €49, K = €50, r = 5%, σ = 20%, μ = 13%, T = 20 weeks = 20/52 years.

The Black-Scholes value of the 20-week European call is:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

with:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Using S_0 = 49, K = 50, r = 0.05, σ = 0.20, and T = 20/52:

$$d_1 \approx 0.0542, \quad d_2 \approx -0.0698$$

so:

$$C \approx 49\, N(0.0542) - 50\, e^{-0.05(20/52)}\, N(-0.0698) \approx 2.4005$$

Thus the Black-Scholes value is approximately:

$$2.4005 \times 100{,}000 = 240{,}053$$

Since the option was sold for €300,000, the initial apparent profit is:

$$300{,}000 - 240{,}053 = 59{,}947$$

Delta Hedging Procedure

To delta hedge the short call, each week the call delta is computed as:

$$\Delta_t = N(d_1)$$

and the number of shares held is adjusted to $100{,}000\Delta_t$. The weekly Excel table tracks: week, stock price, delta, shares purchased, shares held, cost of shares bought, interest cost, and cumulative cost. The stock price is simulated each week using a lognormal process with drift μ = 13% and volatility 20%. Interest accrues each week at r/52.

Delta Hedging Table (Full 20 Weeks)

Wk	Price	Δ	Shares Purch.	Shares Held	Cost Shares	Interest	Cum. Cost
0	49.00	0.5216	52,160	52,160	2,555,863	0	2,555,863
1	48.66	0.4945	-2,715	49,445	-132,121	2,458	2,426,200
2	46.90	0.3678	-12,666	36,779	-594,040	2,333	1,834,493
3	50.16	0.5904	22,260	59,040	1,116,645	1,764	2,952,902
4	51.55	0.6806	9,021	68,061	465,058	2,839	3,420,799
5	52.20	0.7219	4,126	72,188	215,388	3,289	3,639,476
6	52.18	0.7232	129	72,316	6,710	3,499	3,649,686
7	53.05	0.7784	5,527	77,843	293,182	3,509	3,946,376
8	53.00	0.7805	210	78,053	11,121	3,795	3,961,292

9	52.65	0.7647	-1,584	76,469	-83,385	3,809	3,881,716
10	52.48	0.7597	-499	75,969	-26,203	3,732	3,859,246
11	51.84	0.7189	-4,075	71,895	-211,229	3,711	3,651,727
12	51.69	0.7127	-628	71,267	-32,465	3,511	3,622,773
13	52.94	0.8175	10,488	81,754	555,182	3,483	4,181,438
14	53.42	0.8625	4,492	86,247	239,963	4,021	4,425,421
15	53.83	0.9028	4,031	90,278	216,988	4,255	4,646,664
16	56.85	0.9920	8,925	99,203	507,361	4,468	5,158,493
17	55.75	0.9906	-140	99,063	-7,801	4,960	5,155,653
18	57.63	0.9999	925	99,989	53,337	4,957	5,213,947
19	59.30	1.0000	11	100,000	662	5,013	5,219,623
20	58.01	1.0000	0	100,000	0	5,019	5,224,642

P&L Computation

At expiry the stock is at €58.01. The call finishes in the money, so the option holder exercises. The payoff is:

$$(58.01 - 50) \times 100{,}000 = 801{,}000$$

The hedger sells the 100,000 shares at €58.01 for €5,801,000. The cumulative hedging cost from the table is €5,224,642. Without hedging, the profit is simply the premium received minus the option payoff:

$$300{,}000 - 801{,}000 = -501{,}000$$

With hedging, the total profit is:

$$300{,}000 + 5{,}801{,}000 - 5{,}224{,}642 - 801{,}000 = 75{,}358$$

Monte Carlo Validation (500 paths)

Metric	Value
Average hedged profit	€53,408
Average unhedged profit	−€59,497
Theoretical edge (BS mispricing)	€59,947

Comments

The results show that the short call position is extremely risky if left unhedged. Delta hedging reduces this exposure substantially and turns the position into a positive-profit trade in this example. Three distinct effects are at play:

First, the pricing edge: we sold the option for €300,000 against a theoretical value of €240,053, pocketing an initial advantage of €59,947. This is the main source of profit under hedging.

Second, the stock drift: the underlying follows a lognormal process with $\mu = 13\%$, which causes the stock to trend upward on average. This means the call frequently finishes in the money, making the unhedged position a losing proposition on average (−€59,497 across 500 simulations).

Third, discrete hedging error: because we rebalance weekly instead of continuously, the hedge is imperfect. Gamma exposure between rebalancing dates introduces variability around the theoretical profit. Over 500 paths, the average hedged profit of €53,408 falls slightly short of the €59,947 edge, and this gap is attributable to the cost of discrete rebalancing.

Question 10: From Binomial Trees to Black-Scholes

$$\Delta t = \frac{T}{n}$$

As $n \to \infty$, $\Delta t \to 0$ and the binomial tree will appear continuous.

At each moment the stock moves $up = e^{\sigma\sqrt{\Delta t}}$ or $down = e^{-\sigma\sqrt{\Delta t}}$

with $\Delta t \approx 0$ this steps become really small.

$$\text{Risk neutral probability} = \frac{e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}} = \frac{e^{r\Delta t} - d}{u - d}$$

$S_T = S_0 \cdot u^j d^{n-j}$ where j is the n^o of upward steps.

$$\text{Call payoff} = \max(S_T - K, 0) = \max(S_0 u^j d^{n-j} - K, 0)$$

Since each step is independent and the probability of an up move is constant, the number of j moves follows a Binomial distribution.

$$P(j \text{ ups}) = \binom{n}{j} p^j (1-p)^{n-j}$$ so the expected payoff is

$$\sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j} \max(S_0 u^j d^{n-j} - K, 0)$$

$$C_0 = e^{-rT} \cdot \left(\sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j} \max(S_0 u^j d^{n-j} - K, 0) \right)$$

$$S_T = S_0 u^j d^{n-j}$$

$$\ln S_T = \ln S_0 + j \ln u + (n-j) \ln d$$

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}$$

$$\ln S_T = \ln S_0 + (2j - n)\sigma\sqrt{\Delta t}$$

as $n \to \infty$ Binomial $\approx$ Normal

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

$$e^{r\Delta t} \approx 1 + r\Delta t, \quad e^{\sigma\sqrt{\Delta t}} \approx 1 - \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t$$

$$e^{-\sigma\sqrt{\Delta t}} \approx 1 - \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t$$

$$p \approx \frac{\sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t}{2\sigma\sqrt{\Delta t}} \approx \frac{1}{2} + \frac{r - \frac{1}{2}\sigma^2}{2\sigma}\sqrt{\Delta t}$$

$$p \approx \frac{1}{2} + \frac{r - \frac{1}{2}\sigma^2}{2\sigma} \cdot \sqrt{\Delta t}$$

$$\ln S_T \sim N\left(\ln S_0 + (r - \frac{1}{2}\sigma^2)T, \sigma^2 T\right)$$

$$S_T = S_0 \exp\left[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\, z\right], \quad z \sim N(0,1)$$

$n \to \infty$

$$C_0 = e^{-rT} \hat{E}[\max(S_T - K, 0)]$$

$$C_0 = e^{-rT} \tilde{E}^{Q} [\max(S_T - K, 0)]$$

Which is the startpoint of the Black-Scholes-Merton Result in page 351 of the tenth edition of the Hull book, since both the $n \to \infty$ steps binomial tree and Black-Scholes model go through the same equation under the same assumption for a non-dividend paying stock, they must be equal.